Свойства степени

$$a^m = \underbrace{a \cdot a \cdot \ldots \cdot a}_{m}$$

$a^n > 0$

1. $a^0 = 1$
2. $a^1 = a$
3. $a^m \cdot a^n = a^{m+n}$
4. $a^m : a^n = a^{m-n}$
5. $a^{-m} = \frac{1}{a^m}$ ✓
6. $a^m \cdot b^m = (a \cdot b)^m$ ✓
7. $a^m : b^m = \left(\frac{a}{b}\right)^m$
8. $a^{\frac{n}{k}} = \sqrt[k]{a^n}$

$3^{\frac{4}{3}} = \sqrt[3]{3^4} =$

$\sqrt{27} \cdot \sqrt{33} \Rightarrow$ $\sqrt[3]{81}$

$27^{1/2} \cdot 33^{1/2} = (27 \cdot 33)^{1/2}$

9. $(a^m)^n = a^{m \cdot n}$

$$\left(\left(3^{\frac{1}{27}}\right)^2\right)^{81} =$$

$$\frac{3^{-10} \cdot 3^5}{3^{-7}} = \frac{3^{-10+5}}{3^{-7}} =$$

$$\frac{3^{-5}}{3^{-7}} = 3^{-5} \cdot 3^7 = 3^{-5+7} =$$

$$3^2 = 9$$

$$3^{-5-(-7)} = 3^{-5+7} = 3^2 = 9$$

$$\frac{3^{-4} \cdot 3^2}{3^{-7}} = 243$$

$$\frac{8^2}{2^2} : 2^4 = \frac{(2^3)^2}{2^2} : 2^4 =$$

$$\frac{2^6}{2^2 \cdot 2^4} = \frac{2^6}{2^6} =$$

1

$$\frac{64}{4} : 16 = \frac{4}{4} = 1$$

$$\frac{4^7}{2^7} : 2^3 = \frac{(2^2)^7}{2^7} : 2^3 = \frac{2^{14}}{2^7} : 2^3$$

$$= \frac{2^{14}}{2^7 \cdot 2^3} = \frac{2^{14}}{2^{10}} =$$

$$14 - 10$$

$$2^4$$

$$\frac{5^7 \cdot 2^{10}}{10^6} = \frac{5^7 \cdot 2^{10}}{(5 \cdot 2)^6} = \frac{5^7 \cdot 2^{10}}{5^6 \cdot 2^6} =$$

$$\frac{5^7 \cdot 2^{10}}{5^6 \cdot 2^6} = 5^{7-6} \cdot 2^{10-6} =$$

$$5^1 \cdot 2^4 = 5 \cdot 16 = 80$$

$$\frac{2^6 \cdot 3^8}{6^5} = \frac{2^6 \cdot 3^8}{(2 \cdot 3)^5} = \frac{2^6 \cdot 3^8}{2^5 \cdot 3^5} =$$

$$= 2^1 \cdot 3^3 = 2 \cdot 27 = 54$$

$$\frac{4^{3,5} \cdot 5^{2,5}}{20^{1,5}}$$

$$\left(\frac{9^{\frac{1}{3}} \cdot 9^{\frac{1}{4}}}{\sqrt[12]{9}}\right)^3 = \left(\frac{9^{\frac{1}{3}+\frac{1}{4}}}{\sqrt[12]{9}}\right)^3$$

$$\left(\frac{9^{\frac{7}{12}}}{9^{\frac{1}{12}}}\right)^3 =$$

$$\frac{7}{12} - \frac{1}{12} = \frac{7-1}{12} = \frac{6}{12} = \frac{1}{2}$$

$$\left(9^{\frac{1}{2}}\right)^3 = \left(\sqrt{9}\right)^3 = 3^3 =$$

$$27$$

$$\left(\frac{5^{\frac{1}{2}} \cdot 5^{\frac{1}{3}}}{\sqrt[6]{5}}\right)^3 = \left(\frac{5^{\frac{5}{6}}}{5^{\frac{1}{6}}}\right)^3 =$$

$$\left(5^{\frac{4}{6}}\right)^3 =$$

$$5^{\frac{4}{2}} = 5^2 = 25$$

Задача 3,5 из 2/3